

Exercise 1-B (Summer 2026)

6.0 VU Advanced Database Systems

Information

Exercise Sheets

This exercise consists of **three exercise sheets** which will be available in TUWEL on:

- Exercise Sheet 1-A (PL/pgSQL): March 2nd
- Exercise Sheet 1-B (Datalog): March 4th
- Exercise Sheet 1-C (Common Table Expressions): March 6th

General

Work through **all three exercise sheets** and compile your answers in a **single report**. Submit the report and your source files as a **single ZIP (.zip)** file on TUWEL. The ZIP (.zip) file must contain:

- one PDF (.pdf) file,
- SQL (.sql) files containing your PostgreSQL commands (per exercise in sheet),
- Datalog (.dl/.rls) files containing your Datalog commands (per exercise in sheet).

SQL/Datalog file naming format: ex-⟨Exercise Sheet⟩-⟨Exercise Number⟩.⟨Question Number⟩.{sql/dl/rls}

Submit the ZIP (.zip) file (max. 20 MB) and register for an exercise interview. **You can only receive points for this exercise if you attend the interview.** We expect you to explain your work in your report, i.e., it is not enough to only write the final answers. Show how you arrived at your answers and, where applicable, discuss your results. **We will not accept any handwritten reports!**

This exercise sheet involves working on a PostgreSQL DBMS and Datalog systems. For these exercises we provide you with web tools. For this particular exercise sheet you will use the PostgreSQL web tool, which is described in detail below.

Deadline

at latest **March 24th 9:00** Upload your submission on TUWEL
Do not forget! ⇒ Register for an exercise interview in TUWEL

Exercise Interviews

During the exercise interview, the correctness of your solution as well as your understanding of the underlying concepts will be assessed.

The scoring of your submission is primarily based on your performance at the exercise interview. Therefore it is possible (in extreme cases) to get 0 points even though the submitted solution was technically correct.

Please be punctual for your exercise interview. Otherwise we cannot guarantee that your full solution can be graded in your assigned time slot. Remember to have your student id with you for the exercise interview. It is not possible to score your solution without an id.

TUWEL Forum

You can use the course forum on TUWEL for clarifying questions regarding the exercise sheets. Please do not post your solutions (even partial) on the forum.

Datalog environment

For this exercise, we provide a web tool with a locally running Nemo Datalog engine. You can access the tool at <https://dbai.tuwien.ac.at/adbs/>. On that page, click the *Open Datalog Tool* button, to launch the Datalog interface, as shown in Figure 1.

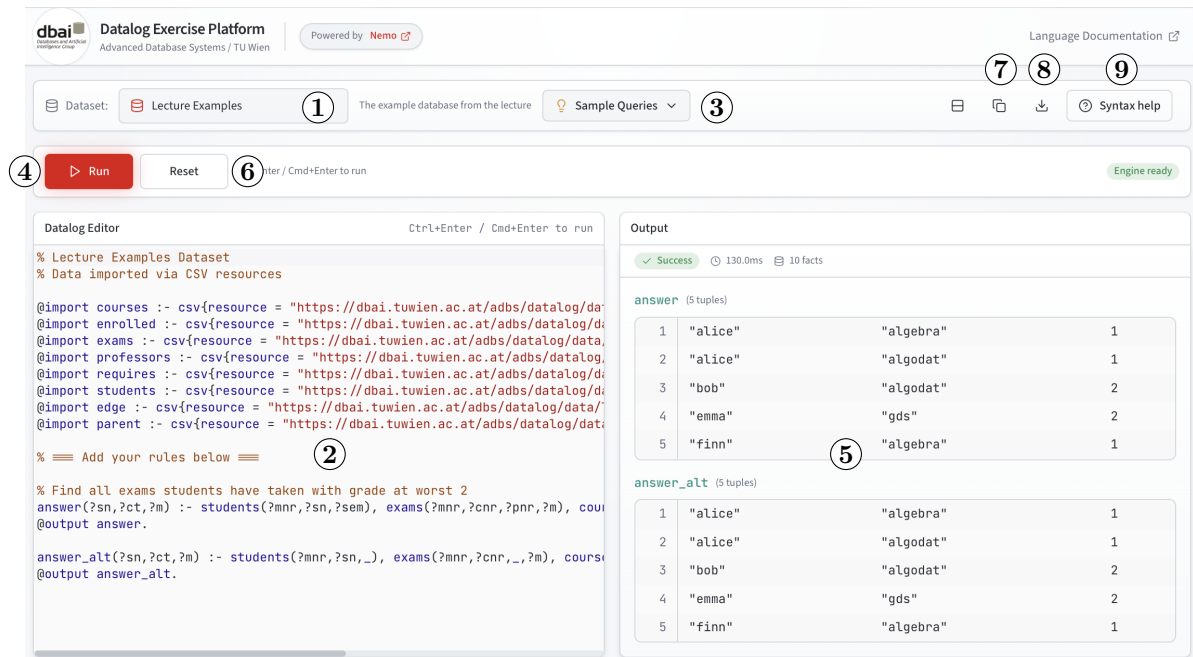


Figure 1: The Datalog Tool

- ① You can use this drop-down menu to load one of the listed datasets.
- ② Once a dataset is loaded, the import statements for the corresponding CSV files will automatically appear in this editor. You can write and test your Datalog programmes there.
- ③ In some cases, sample programmes are provided for exploration. You can select a sample programme from this drop-down menu, and it will appear in ②.
- ④ Click this Run button to execute your programme written in the editor ②.
- ⑤ You can see the outputs of your executed programme here.
- ⑥ If you wish to start over, this Reset button will clear all changes in the tool and datasets. You can reload a dataset from the drop-down menu ① and start over.
- ⑦ This button will allow you to copy your programme currently in the editor.
- ⑧ This button will allow you to download your programmes in the editor into a .rls file.
- ⑨ Click this button to view Datalog syntax help and examples.

Exercises

In exercises 3 and 4 you will familiarize yourself with Datalog to understand its syntax, semantics, and evaluation strategies, and its role as a declarative query language in database systems. The lecture on Datalog introduced the main concepts and principles of the language, which are meant to provide helpful guidance as you work through these exercises.

Exercise 3 (Datalog Basics) [3 points]

We provide a graph of cities with connecting flights, which is also available as the *Flight Database* in the Datalog web tool, shown below:

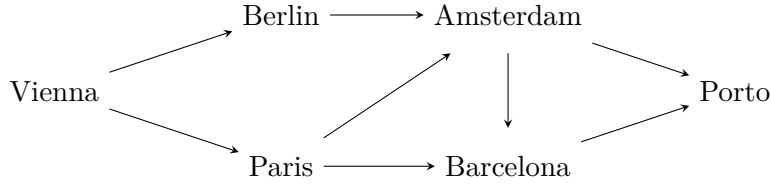


Figure 2: Graph of connected flights

```

flight("Vienna", "Berlin"), flight("Vienna", "Paris"), flight("Berlin",
"Amsterdam"), flight("Paris", "Amsterdam"), flight("Paris", "Barcelona"),
flight("Amsterdam", "Barcelona"), flight("Barcelona", "Porto"), flight("Amsterdam",
"Porto")

```

Consider the following non-linear Datalog programme that computes connecting routes between cities based on the available direct flights:

```

route(?x, ?y) :- flight(?x, ?y)
route(?x, ?y) :- route(?x, ?z), route(?z, ?y).

```

Provide the optimised semi-naive evaluation for this programme along with corresponding Δ_{route}^i , $temp_{route}^i$ and $route^i$ at each step.

Exercise 4 (Datalog with Negation) [2 points]

For this exercise, use the “Employee Database” dataset in the Datalog web tool. It contains the following two relations:

```

employee(emp_id, first_name, last_name, job_id, salary, manager_id, dept_id),
department(dept_id, name, manager_id, location)

```

1. Translate the following relational algebra query that retrieves all employees who manage at least one person, but are not the head of any department and who do not manage anyone working in the Executive department. to non-recursive Datalog with negation ($nr - Datalog^{\neg}$).

```

S =  $\pi_{manager\_id}(employee)$ 
DeptHeads =  $\pi_{manager\_id}(department)$ 
NonDeptHeads =  $employee \bowtie_{emp\_id = manager\_id} (S - DeptHeads)$ 
ExecDept =  $\pi_{dept\_id}(\sigma_{name = 'Executive'}(department))$ 
ManagersExec =  $\pi_{manager\_id}(employee \bowtie_{dept\_id = ExecDept.dept\_id} ExecDept)$ 
Result =  $NonDeptHeads - (NonDeptHeads \bowtie_{emp\_id = ManagersExec.manager\_id} ManagersExec)$ 

```

2. Compute the strata for each predicate. Explain the evaluation of the programme using the computed strata.